

VISVESVARAYA TECHNOLOGICAL UNIVERSITY**“JNANA SANGAMA”, BELAGAVI-590018**

A PROJECT REPORT ON

**“R-PUF: A RESILIENT PHYSICALLY UNCLONABLE FUNCTION FOR
HARDWARE AUTHENTICATION”**

Submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING**In****ELECTRONICS AND COMMUNICATION ENGINEERING**

For the academic year 2023 – 2024

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on

**“R-PUF: A RESILIENT PHYSICALLY UNCLONABLE FUNCTION FOR
HARDWARE AUTHENTICATION”**

This is to certify that the project entitled **“R-PUF: A RESILIENT PHYSICALLY UNCLONABLE FUNCTION FOR HARDWARE AUTHENTICATION”** is a bonafide work carried out by Lakshmi Manoj (1JS20EC040), Nikhil Thevar (1JS20EC052), Sanjana James (1JS20EC076), in partial fulfilment for the award of degree of **Bachelor of Engineering in Electronics & Communication Engineering** of **Visvesvaraya Technological University**, Belagavi during the year 2022-23. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect to project work prescribed for the Bachelor of Engineering degree.

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DECLARATION

We **Lakshmi Manoj, Nikhil Thevar & Sanjana James**, of Bachelor of Engineering in Electronics and Communication, JSS Academy of Technical Education, Bengaluru, hereby declare that the project entitled “**R-PUF:A RESILIENT PHYSICALLY UNCLONABLE FUNCTION FOR HARDWARE AUTHENTICATION**” has been carried out independently by us at JSS Academy of Technical Education, Bengaluru, under the guidance **Dr. Thejaswini P**, Associate Professor, Department of Electronics and Communication JSSATEB, Bengaluru.

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ACKNOWLEDGEMENT

No work is complete without due recognition being given to people who made it possible. This project is no exception. We would like to place on record our profound gratitude for those who have helped us complete this project.

We thank his holiness **Jagadguru Sri Shivarathri Deshikendra Mahaswamiji** for his blessings to this venture.

We are grateful to **Dr. Bhimasen Soragaon**, Principal and **Dr. P M Shivakumaraswamy**, HOD, Department of Electronics and Communication Engineering, for their support and encouragement towards the presentation of this project.

We would like to thank **Dr. Usha S M** and **Dr. Mallikarjunaswamy S**, Project Coordinators for their able guidance. With their help and suggestions, the completion of this project has been easier for us.

We would like to thank **Dr. Thejaswini P**, Project Guide whose modesty, adaptability and cheerful disposition are laudable. **She** inculcated in us an enduring interest to delve further into the topic.

We also express our deep sense of gratitude to all the **Teaching and non-teaching staff** of Electronics and Communication Engineering Department for their valuable guidance.

We would like to express our sincere thanks to our **family and friends** who, in their own ways helped us to finish this project.

ABSTRACT

In the modern era, integrated circuits (ICs) face escalating threats including sophisticated attacks, third-party foundry compromises, and reverse engineering. These vulnerabilities underscore the necessity for robust security solutions to safeguard sensitive hardware. Physical Unclonable Functions (PUFs) emerge as essential hardware security primitives, generating unique identifiers based on inherent physical variations. However, the rising threat of Machine Learning (ML) attacks poses a significant challenge to the security of traditional PUF implementations. As attackers increasingly employ ML techniques to predict and replicate PUF responses, the vulnerability of these systems escalates, potentially undermining their reliability and security. In this project, we propose a novel approach to fortify PUFs against ML attacks through the implementation of Elliptic Curve Cryptography (ECC) encryption. By leveraging ECC's strong security features and lower computational requirements, we aim to encrypt the communication and data involving PUFs, ensuring that even if adversaries analyze intercepted data using ML techniques, they are thwarted from deciphering the unique identifiers or exploiting them for unauthorized access or cloning



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Project Work**Course Outcomes****Sub Code: 18ECP83**

COs	PROJECT WORK	Blooms Level
C411.1	Apply the conceptual interpretation of the mathematical principles, knowledge and skills in the field of Electronics and Communication Engineering to formulate a real time problem.	L3, L4
C411.2	Analyze the formulated engineering problem and apply appropriate Engineering techniques with suitable design process.	L4
C411.3	Develop creative solutions with appropriate state-of-the-art techniques/tools.	L4
C411.4	Test and validate the solution of the developed prototype against the design specifications.	L4
C411.5	Demonstrate the knowledge of professional ethics in documentation and presentation of the work.	L4

This Project maps to the following Program Outcomes:

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C411.1	3					2	2	2	2	2	3	2
C411.2		3	3									
C411.3			3	3	3							
C411.4				3	3			2				
C411.5						2	2		2	2	3	2
C411	3	3	3	3	3	2	2	2	2	2	3	2

CO/PSO	PSO1	PSO2	PSO3
C411.1	3	-	3
C411.2	3	3	3
C411.3	3	3	3
C411.4	3	3	3
C411.5	3	-	-
C411	3	3	3

COs	POs	Level	Justification
C411.1	PO1	3	The Project problem formulation needs application of mathematics, science & engineering fundamentals.
	PO6	2	The project work address societal needs, health, safety & legal issues with responsibilities relevant to professional engineering and practice
	PO7	2	The project work demonstrates the engineering knowledge needed for sustainable development.
	PO8	2	The project work applies ethical principles, professional responsibility in the process of implementation, report preparation & presentation.
	PO9	2	The project work is carried out by group of two or four students which involves individual and teamwork.
	PO10	2	The project work is to be presented orally and exhibited in the exhibition through which the work is communicated effectively. The project work is documented in the report form based on the instructions and corrections mentioned by the faculty.
	PO11	3	The project work demonstrates project management to execute effectively by the students (Individual & team work). The project work also addresses the financial aspects in its implementation.
	PO12	2	The project work prepares the students for independent and lifelong learning in the context of technological change.
C411.2	PO2	3	The project work identifies the issues and formulates the problem through review of research literature. Analyze the engineering problem using principles of mathematics, science & engineering
	PO3	3	The project work has to design or develop solutions for the problems formulated with consideration for health and safety, cultural, societal and environmental needs.
C411.3	PO3	3	The project work has to design or develop solutions for the problems formulated with consideration for health and safety, cultural, societal and environmental needs.
	PO4	3	The project work provides valid conclusion using research-based knowledge design of experiments analysis and interpretation of data.
	PO5	3	The project work needs application of appropriate techniques resources and modern engineering, IT tools for prediction and modeling of the designed or developed system.
C411.4	PO4	3	The project work provides valid conclusion using research based knowledge design of experiments analysis and interpretation of data.
	PO5	3	The project work needs application of appropriate techniques resources and modern engineering, IT tools for prediction and modeling of the designed or developed system.
	PO8	2	The project work applies ethical principles, professional responsibility in the process of implementation, report preparation & presentation.

C411.5	PO6	2	The project work in its documentation & presentation address societal needs, health, safety & legal issues with responsibilities relevant to professional engineering and practice
	PO7	2	The project work has to document & present the engineering knowledge needed for sustainable development.
	PO9	2	The project work is carried out by group of two or four students which involves individual and teamwork in preparation of documentation & presentation.
	PO10	2	The project work is to be presented orally and exhibited in the exhibition, through which the work is communicated effectively. The project work is documented in the form of report based on the instructions and corrections mentioned by the faculty.
	PO11	3	The project work demonstrates project management to execute effectively by the students (Individual & team work). The project work also addresses the financial aspects in its implementation which are documented in the report and presented.
	PO12	2	The process of project work documentation & presentation prepares the students for independent and lifelong learning in the context of technological change.

CO - PSO Justification

COs	PSOs	Level	Justification.
C411.1	PSO1	3	The project work aims to solve engineering problem of electronics communication in VLSI Design, Embedded Systems, Communication Engineering, Power Electronics and Control Systems.
	PSO3	3	The project work applies mathematical principles, knowledge and skills of Electronics & Communication Engineering in the designing, realization and validation of various electronic systems.
C411.2	PSO1	3	The project work solves engineering problems by using appropriate engineering technique with suitable design process in the field of Electronics & Communication Engineering.
	PSO2	3	To analyse the formulated engineering problem, suitable programming skills and high-level languages are necessary.
	PSO3	3	The design of project work needs analysis of the formulated engineering problems.
C411.3	PSO1	3	To solve engineering problem in the field of electronics and communication, state- of- the-art, modern tools or technique has to be used in the project work.
	PSO2	3	Programming skills and assembly high level language will help in solving engineering problems.
	PSO3	3	To design, realize and validate electronics system the programming skills using assembly and high-level languages are essential.

C411.4	PSO1	3	The designed electronics system has to be tested and validated before any real time implementation of the product.
	PSO2	3	Testing and validation is carried out with programming skills in assembly and high level languages.
	PSO3	3	Electronics system is designed and validated before the product development.
C411.5	PSO1	3	The project work solves a problem in Electronics & Communication Engineering needs to be documented and presented for sustainable development of the project work.

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CHAPTER 1

INTRODUCTION

In the ever-evolving landscape of cybersecurity, safeguarding hardware authentication mechanisms against sophisticated attacks is paramount. As technology continues to advance, adversaries are employing increasingly complex techniques to compromise systems and gain unauthorized access, posing significant threats to critical infrastructure, secure communication networks, and sensitive data. Consequently, robust hardware authentication mechanisms have become crucial for ensuring the integrity and authenticity of electronic devices, particularly in applications where security breaches can have severe consequences.

Among the arsenal of cryptographic tools, the Physical Unclonable Function (PUF) has emerged as a cornerstone technology, offering unique hardware-derived identifiers for authentication. PUFs leverage the inherent manufacturing process variations in integrated circuits to generate unique, unclonable identifiers or cryptographic keys. These identifiers, derived from the physical characteristics of the hardware itself, are virtually impossible to replicate or predict, making them highly attractive for applications such as device authentication, key generation, and secure storage.

1.1 Background

i. Physical Unclonable Functions (PUFs)



Fig 1a Physical Unclonable Function

PUFs can be categorized into various types based on their underlying principles and implementation methods. One of the most widely studied and implemented PUF architectures is the Arbiter PUF (APUF). APUFs exploit the random delay variations in identically designed circuit paths to generate unique responses. These responses are determined by an arbiter circuit that selects the faster signal propagation path, resulting in a binary output.

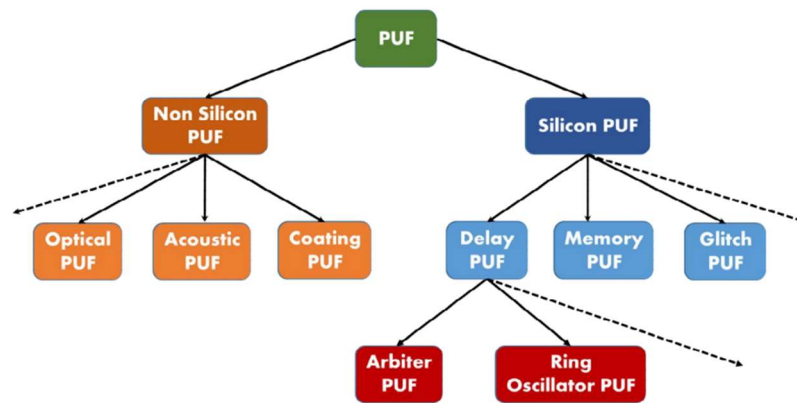


Fig 1b Types Physical Unclonable Function

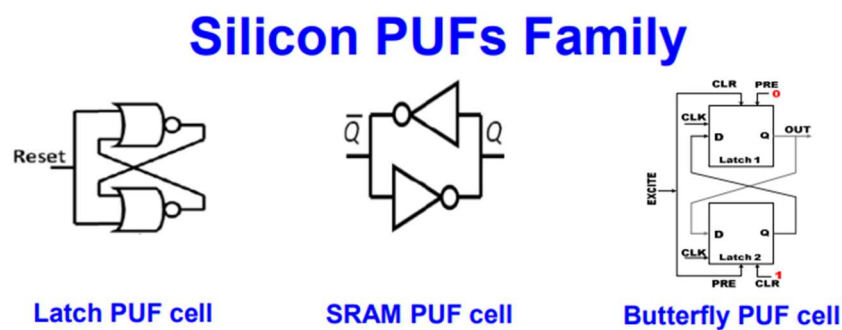


Fig 1c Types of Silicon PUF

The APUF architecture typically consists of two parallel signal paths, each comprising multiple stages of multiplexers and interconnects. When a challenge (a sequence of bits) is applied to the multiplexers, it selects the path taken by the signals. Due to manufacturing variations, the delays in these paths are slightly different, leading to a delay difference at the arbiter circuit. The arbiter acts as a race condition detector, producing a '1' or '0' output based on which signal arrives first.

Other notable PUF types include Ring Oscillator PUFs (RO-PUFs), which utilize the frequency differences in identically designed ring oscillators to generate unique signatures. In RO-PUFs, the frequency variations in the oscillators are caused by manufacturing process variations, and these frequencies are measured and used as the PUF response.

SRAM PUFs, another widely explored PUF design, leverage the manufacturing variations in SRAM memory cells to generate unique signatures. During power-up, the SRAM cells settle into random '0' or '1' states due to these variations, and this initial state can be used as the PUF response.

Additionally, researchers have explored lightweight PUF designs tailored for resource-constrained devices, such as IoT (Internet of Things) and embedded systems. These lightweight PUFs aim to provide security features with minimal hardware overhead, making them suitable for low-cost and low-power applications.

While PUFs offer several advantages, including low-cost implementation and resistance to physical cloning, they are not immune to various attacks, particularly Machine Learning (ML) attacks. ML attacks have emerged as a significant threat to the security and reliability of PUF-based authentication systems, necessitating robust countermeasures and security enhancements.

ii. Machine Learning Attacks on PUFs

ML attacks on PUFs have been extensively studied and demonstrated by researchers, highlighting the vulnerability of these hardware security primitives to advanced computational techniques. These attacks leverage various machine learning algorithms, such as logistic regression, evolution strategies, and computational intelligence algorithms, to model the behavior of a PUF by analyzing a set of Challenge-Response Pairs (CRPs).

The goal of ML attacks is to train a machine learning model on a subset of CRPs obtained from a PUF, enabling the attacker to accurately predict the responses for new, unseen challenges. By successfully modeling the PUF's behavior, the attacker can effectively break the security of the PUF-based authentication system, rendering it vulnerable to unauthorized access or spoofing attacks.

Several PUF constructions, including Arbiter PUFs and their variants, have been shown to be susceptible to ML attacks, particularly when the attacker has access to a large number of CRPs. This vulnerability highlights the need for robust countermeasures and security enhancements to protect PUFs from such attacks and ensure the integrity of hardware authentication mechanisms.

Researchers have proposed various countermeasures to mitigate ML attacks on PUFs, including the introduction of non-linearity, the use of lightweight PUF designs, and the incorporation of additional security measures. However, many of these approaches have their own limitations or trade-offs, such as increased hardware complexity, reduced reliability, or performance overhead.

iii. Elliptic Curve Cryptography (ECC)

One promising approach to mitigate ML attacks on PUFs is to incorporate advanced cryptographic techniques, such as Elliptic Curve Cryptography (ECC). ECC is a type of public-key cryptography based on the algebraic structure of elliptic curves over finite fields. It offers several advantages over traditional cryptosystems like RSA, including smaller key sizes, higher computational efficiency, and reduced memory and bandwidth requirements.

The security of ECC relies on the computational complexity of the Elliptic Curve Discrete Logarithm Problem (ECDLP), which is believed to be significantly harder to solve than the integer factorization problem underlying RSA cryptosystems. This makes ECC an attractive choice for applications requiring strong security with limited computational resources, such as embedded systems and IoT devices.

ECC has been widely adopted in various security protocols and applications, including secure communication protocols (e.g., TLS, IPsec), digital signatures, and key exchange mechanisms. Its efficiency and security properties make it a suitable candidate for integration with PUFs to enhance the overall security of hardware authentication systems.

By leveraging the mathematical properties of elliptic curves, ECC can be employed to encrypt the CRPs generated by an Arbiter PUF, effectively obscuring the challenge-response relationship and preventing adversaries from obtaining the necessary data to mount successful ML attacks. This encryption process ensures the confidentiality and integrity of the CRPs, safeguarding them against unauthorized access or manipulation.

iv. ECC-Encrypted Arbiter PUFs for Hardware Authentication

This paper focuses on the design, implementation, and analysis of an ECC-encrypted Arbiter PUF for robust hardware authentication. By integrating ECC encryption with the Arbiter PUF architecture, the proposed solution aims to mitigate ML attacks while preserving the inherent advantages of PUFs, such as low-cost implementation and resistance to physical cloning.

The key idea behind the ECC-encrypted Arbiter PUF is to encrypt the CRPs generated by the Arbiter PUF using an ECC-based encryption scheme. This encryption process effectively conceals the challenge-response relationship, preventing adversaries from obtaining the necessary data to train ML models and predict responses for unseen challenges.

Furthermore, the ECC-encrypted Arbiter PUF leverages the mathematical properties of elliptic curves to ensure the confidentiality and integrity of the encrypted CRPs, safeguarding them against unauthorized access or manipulation. By combining the strengths of Arbiter PUFs and ECC, the proposed solution offers a robust and secure hardware authentication mechanism that addresses the vulnerability of Arbiter PUFs to ML attacks while maintaining the advantages of PUFs, such as low cost and resistance to physical cloning.

The proposed ECC-encrypted Arbiter PUF architecture incorporates several key components, including the Arbiter PUF core, an ECC encryption module, and a secure key storage mechanism. The Arbiter PUF core generates the CRPs based on the applied challenges, while the ECC encryption module utilizes a pre-shared secret key to encrypt the CRPs before they are transmitted or stored. The secure key storage mechanism ensures the confidentiality of the secret key used for encryption, protecting it from unauthorized access or tampering.

To further enhance security, the ECC-encrypted Arbiter PUF can be integrated with additional security measures, such as challenge-response protocols, mutual authentication mechanisms, and secure key exchange protocols. These measures provide added layers of protection against potential attacks, ensuring the overall robustness and reliability of the hardware authentication system.

By addressing the vulnerability of Arbiter PUFs to ML attacks through the integration of ECC encryption, the proposed solution contributes to the ongoing efforts to strengthen hardware security and protect critical systems from sophisticated threats. The remainder of this paper is structured as follows...

1.1 Problem Statement

We aim to develop a resilient Physically Unclonable Function (PUF) design that effectively counters machine learning attacks, surpassing existing state-of-the-art solutions. Our objective is to create a robust PUF implementation that withstands adversarial machine learning techniques, ensuring enhanced security and reliability

1.2 Objectives

1. Analyze and evaluate existing machine learning-based PUF attacks.
2. Execute targeted attacks on established State-of-the-Art PUF designs.
3. Design a robust PUF.
4. Validate functionally and verify the effectiveness of the designed PUF against the identified attack methodologies.

CHAPTER 2

LITERATURE SURVEY

1. PI PUF: A Processor-Intrinsic PUF for IoT

The concept of using intrinsic processor characteristics as an entropy source for Physical Unclonable Functions (PUFs) was first proposed by Guajardo et al. in their 2007 work "FPGA Intrinsic PUFs and Their Use for IP Protection". They demonstrated the use of timing characteristics of data paths within a processor to generate unique signatures. However, their approach suffered from limited entropy and vulnerability to machine learning attacks.

Kodych and Levi's 2010 paper "Exploiting Determinism in Computer Processors to Provide Hardware Keys" improved upon Guajardo's work by introducing instruction-level obfuscation techniques. They executed carefully crafted instruction sequences to amplify the timing variations across different processor components, thereby increasing the entropy of the generated PUF responses.

Saraswat et al. further built upon these ideas in their 2013 work "Enhancing Security of Processor-Based Challenge-Response Systems Through Workload Obfuscation". They proposed a multi-round obfuscation mechanism that exploited timing dependencies across multiple pipeline stages, leading to higher entropy and better resistance against machine learning attacks.

More recently, Cui et al. presented a "Branch-Instruction-Based Embedded Processor-Intrinsic Physical Unclonable Function" in 2018. Their design leveraged the timing variations in the execution of branch instructions to generate PUF responses, aiming to provide higher entropy and better resilience against environmental variations.

In 2020, Mehrdad et al. proposed a "Memory-Based Hardware PUF for IoT Security" that utilized the timing characteristics of memory access operations as the entropy source. Their design targeted low-cost implementation and protection against machine learning attacks.

While these works have advanced the field of processor-intrinsic PUFs, challenges remain in terms of ensuring reliability under varying environmental conditions, improving entropy and uniqueness, and developing efficient and secure protocols for key generation, authentication, and secure storage based on PI PUFs.

The "PI PUF: A Processor-Intrinsic PUF for IoT" paper aims to address these challenges by proposing a novel PI PUF design tailored for IoT security, incorporating techniques like multi-round instruction obfuscation, Gray code encoding, and a lightweight authentication protocol with security measures like shared key encryption and XOR random number encryption.

2. Towards Ideal Arbiter PUF Design on Xilinx FPGA: a Practitioner's Perspective

Arbiter PUFs (APUFs) are a widely studied type of delay-based PUF that exploit the random manufacturing variations in the delays of identically designed circuit paths. Majzoobi et al.'s 2008 work "Lightweight Secure PUFs" was one of the earliest implementations of APUFs on FPGAs using programmable delay lines (PDLs). However, their design suffered from non-ideal statistical properties like bias and low uniqueness due to systematic variations introduced by the FPGA CAD tools,

Subsequent works attempted to mitigate these issues by proposing techniques like XOR combinatorial loops (Maiti et al., 2010) and feed-forward arbiters (Majzoobi et al., 2010). While these approaches improved the statistical properties, they came at the cost of increased area overhead and complexity.

Cherubini et al.'s 2011 paper "Advancing Noisiness in Arbiter PUFs" introduced the concept of using multi-bit PDLs to increase the noisiness of the delay paths, thereby enhancing the uniqueness and reliability of APUF responses. However, their implementation on Xilinx FPGAs still suffered from systematic biases due to the FPGA routing architecture.

Gao et al.'s 2015 work "Reverse Engineering Attacks on Public PUF Implementations with Arbitrary and a Priori Unknown Delay Characteristics" highlighted the vulnerability of APUF designs to machine learning attacks, emphasizing the need for bias-free and symmetric delay paths.

The "Towards Ideal APUF Design on Xilinx FPGA" paper builds upon these insights and proposes a novel methodology to overcome the limitations of traditional FPGA CAD flow in implementing high-quality APUFs. By exploiting the Hard Macro feature of Xilinx design flow, the authors design bias-free symmetric delay paths and mitigate systematic biases arising from PDLs, arbiter circuit, and input trigger generation.

Their key contributions include using multi-bit PDL hard macros for upper and lower delay paths, incorporating input registers to reduce trigger signal bias, and using an SR latch-based arbiter to avoid routing skew. This results in superior statistical properties like high uniformity, reliability, and uniqueness compared to previous PDL-based APUF designs on FPGAs.

The proposed methodology enables scalable implementation of large, high-quality APUFs on FPGAs, overcoming the limitations of traditional FPGA CAD flow and addressing the vulnerabilities of previous APUF designs. This work represents a significant step towards practical and secure APUF implementation on FPGAs for various applications.

3. PUF Modeling Attacks: An Introduction and Overview

The concept of modeling attacks against Physical Unclonable Functions (PUFs) was first introduced by Lim et al. in their 2005 work "Extracting Secret Keys from Integrated Circuits". They demonstrated the possibility of using machine learning techniques to model the behavior of PUFs based on a set of challenge-response pairs (CRPs), enabling an attacker to predict responses for unseen challenges.

One of the earliest modeling attacks was proposed by Rührmair et al. in their 2010 paper "Modeling Attacks on Physical Unclonable Functions". They applied logistic regression and evolution strategies to model the behavior of Arbiter PUFs and achieved high prediction accuracies, highlighting the vulnerability of these PUF constructions to modeling attacks.

Subsequent works explored the effectiveness of various machine learning techniques for modeling attacks. Hospoar et al. (2012) used computational intelligence techniques like evolutionary algorithms to model XOR Arbiter PUFs, while Becker (2015) employed an optics algorithm-based attack to model lightweight PUFs.

In response to these attacks, researchers proposed countermeasures to enhance the resilience of PUFs against modeling. Majzoobi et al. (2010) introduced the XOR Arbiter PUF, which combined multiple Arbiter PUFs using an XOR operation to increase non-linearity and improve resistance against modeling attacks.

Gao et al. (2016) analyzed the effectiveness of XORing techniques and proposed a new PUF design called the Interpose PUF, which incorporated non-linearity and tamper-resistance features to thwart modeling attacks.

Nguyen et al. (2019) investigated the vulnerability of lightweight PUFs, which were initially considered more resilient against modeling attacks due to their simplicity and non-linearity. They demonstrated successful modeling attacks on various lightweight PUF constructions, highlighting the need for further research in this area.

The "PUF Modeling Attacks: An Introduction and Overview" paper provides a comprehensive analysis of these modeling attack techniques and the robustness of different PUF constructions against such attacks. It highlights the importance of developing lightweight, robust, and attack-resilient PUF designs for practical deployments, as modeling attacks pose a significant threat to the security of PUF-based systems.

4. FPGA-based Physical Unclonable Functions: A comprehensive overview of theory and architectures

One of the earliest works on implementing Physical Unclonable Functions (PUFs) on FPGAs was by Syssau et al. in their 2006 paper "A Reconfigurable Instruction Cell Array for Trusted Computing". They proposed using the inherent delay variations in the routing fabric of FPGAs to generate unique signatures for device authentication.

Majzoobi et al.'s 2008 work "Lightweight Secure PUFs" presented one of the first implementations of the Arbiter PUF (APUF) on FPGAs using programmable delay lines. However, their design suffered from non-ideal statistical properties due to systematic variations introduced by the FPGA CAD tools.

Maiti et al.'s 2010 paper "A Large Scale Implementation of Controlled Physical Unclonable Functions" introduced the concept of XOR APUFs, which combined multiple APUFs using XOR gates to improve resistance against modeling attacks and enhance uniqueness.

Cherubini et al.'s 2011 work "Advancing Noisiness in Arbiter PUFs" proposed using multi-bit programmable delay lines to increase the noisiness of delay paths in APUFs, leading to improved reliability and uniqueness.

Subsequent works explored other types of PUFs on FPGAs, such as Lao and Parhi's 2011 paper "Obfuscated Circuits for Secure FPGA-based Key Storage", which implemented memory-based PUFs using SRAM and DRAM on FPGAs.

Gao et al.'s 2015 work "Reverse Engineering Attacks on Public PUF Implementations with Arbitrary and a Priori Unknown Delay Characteristics" highlighted the vulnerability of APUF designs to machine learning attacks, emphasizing the need for countermeasures.

Gao et al.'s 2016 paper "Interpose PUF: Interpose Circuit Extension with Integrated Mirror Key-Registers for Low-Cost Secured PUF Scan and Latest Advances" proposed the Interpose PUF, which incorporated non-linearity and tamper-resistance features to thwart modeling attacks.

The "FPGA-based Physical Unclonable Functions: A comprehensive overview of theory and architectures" paper provides a comprehensive review of these works and other developments in the field of FPGA-based PUFs. It discusses the architectures, design considerations, and performance metrics of various delay-based and memory-based PUF implementations on FPGAs. The paper also covers PUF modeling attacks and countermeasures, highlighting the advantages and challenges of using FPGAs for PUF design.

5. Elliptic Curve Cryptography: Algorithms and Implementation Analysis over Coordinate Systems

The foundational work in elliptic curve cryptography (ECC) was the 1985 paper "Use of Elliptic Curves in Cryptography" by Miller and Koblitz, which proposed using elliptic curves for designing public-key cryptosystems.

One of the earliest works analyzing ECC point arithmetic algorithms over different coordinate systems was Lopez and Dahab's 1999 paper "Fast Multiplication on Elliptic Curves over $GF(2^m)$ without Precomputation". They introduced the Lopez-Dahab coordinate system and demonstrated its computational efficiency for point multiplication on binary elliptic curves.

Cohen et al.'s 1998 paper "Analyzing the Elliptic Curve Computation Analytical Treatment of Coordinate Systems" provided a comprehensive analysis of various coordinate systems, including affine, projective, Jacobian, and Chudnovsky, for point arithmetic over prime fields.

Hankerson et al.'s 2004 book "Guide to Elliptic Curve Cryptography" served as a valuable reference, covering ECC algorithms, coordinate systems, and implementation aspects in detail.

Subsequent works focused on optimizations and countermeasures for ECC implementations. Joye and Quisquater's 2003 paper "Hessian Elliptic Curves and Side-Channel Attacks" introduced the Hessian coordinate system and analyzed its resistance against side-channel attacks.

Bernstein and Lange's 2007 paper "Curve25519: New Diffie-Hellman Speed Records" proposed the Curve25519 parameter and presented efficient implementations using the Edwards coordinate system and other optimizations.

Hamburg's 2015 paper "Revisiting the Computational Practicality of Curve25519" further optimized Curve25519 implementations using vector instructions and other techniques.

Works by Hisil et al. (2009) and Renes et al. (2016) explored efficient endomorphisms and curve parameterization techniques to speed up ECC computations.

The "Elliptic Curve Cryptography: Algorithms and Implementation Analysis over Coordinate Systems" paper provides a comprehensive review of these works and other developments in ECC algorithms and implementations across different coordinate systems. It covers the mathematical background, point arithmetic algorithms, optimization techniques, countermeasures against side-channel attacks, and implementation analysis on various platforms, serving as a valuable reference for researchers and practitioners in the field of ECC.

6. Multiple Encryption using ECC and Its Time Complexity Analysis

The concept of combining multiple encryption schemes to enhance security is not new. Goh et al.'s 2003 paper "Analyzing and Revising PGPnet's Conventions" explored the idea of multiple encryption in the context of the PGPnet protocol, which used a two-layer encryption scheme with RSA and IDEA.

With the increasing adoption of Elliptic Curve Cryptography (ECC) due to its efficiency and security advantages, researchers began exploring the use of ECC in hybrid encryption schemes. Libert and Quisquater's 2006 paper "Efficient Signcryption with Key Privacy from Gap Diffie-Hellman Groups" proposed a signcryption scheme that combined ECC with symmetric encryption for efficient authenticated encryption.

Odelu et al.'s 2015 work "Multiple Encryption Using ECC with Multi-Prime RSA and GOST" presented a hybrid encryption scheme that combined ECC with the multi-prime RSA and GOST algorithms, analyzing the security and time complexity of the proposed approach.

Researchers have also explored the use of ECC in conjunction with other cryptographic primitives, such as hash functions and pseudorandom number generators. Naor and Segev's 2009 paper "Public-Key Cryptosystems Resilient to Key Leakage" proposed a leakage-resilient public-key encryption scheme that combined ECC with a hash function and pseudorandom number generator.

In addition to theoretical analyses, several works have focused on the practical implementation and performance evaluation of multiple encryption schemes involving ECC. Amin et al.'s 2014 paper "Hybrid Cryptography: A Combination of ECC and AES" presented a hybrid encryption scheme that combined ECC with the Advanced Encryption Standard (AES) and evaluated its performance on various platforms.

The "Multiple Encryption using ECC and Its Time Complexity Analysis" paper aims to provide a comprehensive analysis of the time complexity and computational overhead associated with employing ECC in multiple encryption schemes. It explores the trade-offs between security and performance, highlighting the importance of considering time complexity alongside cryptographic strength in designing efficient and secure encryption protocols.

By reviewing and building upon these existing works, the paper contributes to the understanding of how the combination of ECC with other encryption algorithms impacts computational time complexity, offering insights into optimizing performance while maintaining robust security measures in multi-layered encryption systems.

CHAPTER 3

METHODOLOGY

The methodology section outlines the detailed steps taken to achieve the project objectives, including the design and implementation of the PUF, the ECC encryption process, the integration of these components, and the evaluation and testing of the system. Each step is crucial for ensuring the security and functionality of the final system.

3.1 PUF Design and Implementation

3.1.1 Selection of PUF Type

For this project, an arbiter PUF was chosen due to its simplicity, ease of implementation, and widespread use in security applications. The arbiter PUF operates by sending an electrical signal through multiple paths with slightly different delays, which are influenced by manufacturing variations.

- Simplicity

The arbiter PUF is known for its straightforward design and relatively simple implementation. Unlike other complex PUF architectures that may require intricate circuitry or sophisticated algorithms, the arbiter PUF relies on basic digital components like multiplexers (MUX) and an arbiter (latch or flip-flop). This simplicity translates into easier hardware implementation and reduced design complexity.

- Ease of Implementation

Implementing an arbiter PUF can be done with minimal overhead in both hardware and software. The design primarily involves creating a sequence of multiplexers arranged in stages, followed by an arbiter to determine the final output. This straightforward approach allows for quick prototyping and testing, especially when using programmable logic devices like FPGAs (Field-Programmable Gate Arrays).

- Widespread Use in Security Applications

The arbiter PUF has been extensively studied and utilized in various security applications. Its popularity stems from its effectiveness in generating unique and reliable responses based on manufacturing variations. This makes it a well-established choice for projects aiming to enhance security through hardware-based solutions.

3.1.2 Arbiter PUF

The concept of an Arbiter PUFs is based on the difference of the delays of two signals, which gets converted to a single bit response. To implement this concept, an Arbiter PUF has multiple connected stages and an arbiter at the end, as shown in Fig.

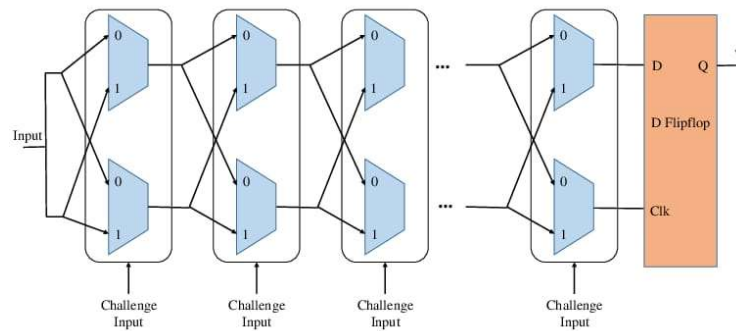


Fig 3.1.2a Arbiter PUF

Both signals propagate on parallel paths with the same layout length through all stages until they reach the arbiter. The arbiter then outputs the response bit depending on which input of the arbiter is triggered first by one of the signals. Every stage has three inputs, two for both signals and one for one bit of the challenge. This challenge bit determines if the signal paths will be crossed in this stage. If the bit is 0, both signals will flow parallel through the stage. If the bit is 1, the stage will swap the signal flow. So the paths for the signals through the stages exclusively depends on the challenge $c = (c_1, c_2, \dots, c_n)$.

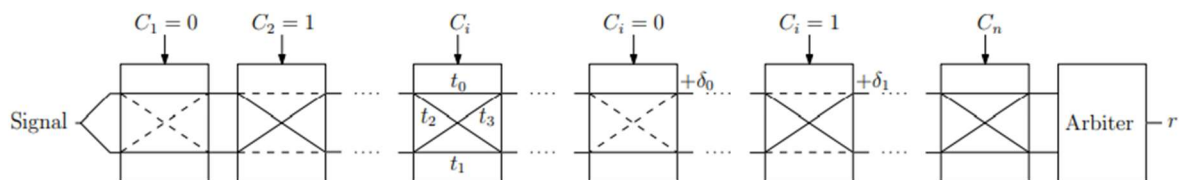


Fig 3.1.2b Arbiter PUF

The challenge bit length must match the number of stages in the Arbiter PUF so every stage gets a challenge bit assigned. This is called the size n of the Arbiter PUF in this work. The circuits of both

paths are laid out to have logical equal length. But, due to the inaccuracies in the manufacturing process of the electronic devices used for the stages, which cannot be influenced by the producer, transit times will differ from device to device. These differences can be seen as the intrinsic unique key of an instance of every Arbiter PUF. One stage for itself has 4 different signal paths:

- top input - top output: t_0
- bottom input - bottom output: t_1
- top input - bottom output: t_2
- bottom input - top output: t_3

Nevertheless, there are just two combinations possible, as the stage input is a Boolean value. Either its top input to top output and bottom input to bottom output or its top input to bottom output and bottom input to top output. These combinations lead to the delay differences $t_0 - t_1 = \delta_0$ and $t_2 - t_3 = \delta_1$ for the possible input 0 or 1 per stage. Additionally, to the delay differences, which remain constant, deviations are occurring during the Arbiter PUF is queried. These deviations are noise and one way to model them is to use two distinct random variables. Both variables are normally distributed with mean zero and can be described as follows:

- At every stage of the Arbiter PUF, additionally to the chosen delay differences δ_0 or δ_1 , normally distributed noise with zero mean and given variance occurs.
- At the arbiter normally distributed noise with zero mean and given variance leads to a different output of the comparison sometimes.

Due to the internal noise, the PUF may return different responses when given the same challenges twice. That is because challenges in combination with the different path delay values can result in a very small difference between the appearing signals on the arbiter. This small difference can be changed by the introduced noise or the arbiter can decide wrong due to its own noise every time the Arbiter PUF is queried and so the final response can be changed. Hence, every challenge has its own stability value and challenges with a lower stability value are called unstable. Whether a challenge is called unstable, depends on its stability value and the limit within a challenge is defined as stable. This limit is not generally defined and can be set as needed. If the stability value is below this limit the challenge is called unstable. These are all parts to evaluate and to specify an Arbiter PUF.

3.2 ECC Encryption Process

3.2.1 Elliptic-curve cryptography

Elliptic-curve cryptography (ECC) is an approach to public-key cryptography based on the algebraic structure of elliptic curves over finite fields. ECC allows smaller keys compared to non-EC cryptography (based on plain Galois fields) to provide equivalent security.

Elliptic curves are applicable for key agreement, digital signatures, pseudo-random generators and other tasks. Indirectly, they can be used for encryption by combining the key agreement with a symmetric encryption scheme. They are also used in several integer factorization algorithms that have applications in cryptography, such as Lenstra elliptic-curve factorization.

An *elliptic curve* is a plane curve over a finite field (rather than the real numbers) which consists of the points satisfying the equation:

$$y^2 = x^3 + ax + b$$

along with a distinguished point at infinity, denoted ∞ . The coordinates here are to be chosen from a fixed finite field of characteristic not equal to 2 or 3, or the curve equation would be somewhat more complicated.

This set of points, together with the group operation of elliptic curves, is an abelian group, with the point at infinity as an identity element. The structure of the group is inherited from the divisor group of the underlying algebraic variety.

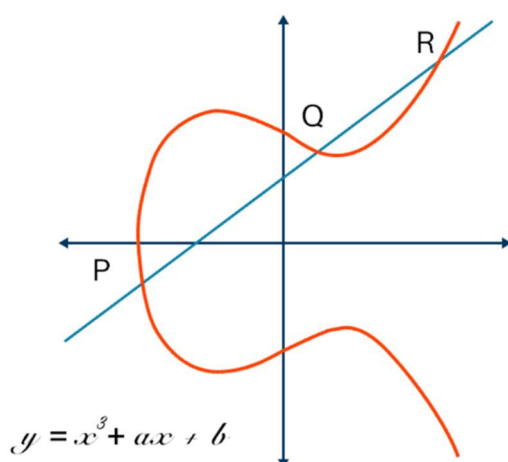


Fig 3.2.1 ECC Curve

3.2.2 ECC Key Generation

Elliptic Curve Cryptography (ECC) is a public-key cryptography technique based on the algebraic structure of elliptic curves over finite fields. The key generation process involves creating a pair of keys: a private key and a corresponding public key.

Steps in ECC Key Generation:

i. Selection of the Elliptic Curve Parameters:

- **Curve Equation:** Choose an elliptic curve defined by an equation of the form
$$y^2 = x^3 + ax + b$$
 over a finite field $\mathbb{F}_p$ (where p is a prime number).
- **Base Point (G):** Select a base point G on the elliptic curve, which is a publicly known point used in the key generation process.

ii. Private Key and Public Key Generation:

- The private key is a randomly chosen integer within the interval $[1, n-1]$, where n is the order of the base point G . The order n is the smallest positive integer such that $nG = O$ (the point at infinity).
- The public key is generated by multiplying the private key with the base point G on the elliptic curve. The result is a point on the elliptic curve and serves as the public key corresponding to the private key.

1. User B key generation:

- Select private key $\rightarrow n_A$ $n_A < n$
- Public key $\rightarrow P_A = n_A * G(x, y)$

2. User A key generation:

- Select private key $\rightarrow n_B$ $n_B < n$
- Public key $\rightarrow P_B = n_B * G(x, y)$

Example for key generation:

- Let $G(x, y) = G(283, 315)$
- Let $n_B = 5$

- Then public key $P_B = n_B * G(x,y)$
 $= 5 * G(283, 315)$
 $= (410, 439)$

3.2.3 ECC Encryption Algorithm

The ECC encryption process for the PUF responses involves:

i. Public Key Distribution: The public key is distributed to all entities that need to encrypt the PUF responses.

ii. Encryption Process: Each PUF response is encrypted using the public key. The encryption process follows these steps:

- Let the puf response be M
- First, we encode the puf response M into a point on elliptical curve P_M
- For the encryption, chose a random positive integer k , such that $k < n$
- The Cipher point will be:

$$C_m = (kG, P_M + kP_B)$$

Example:

- Let the message to be encoded be 'd'
- From the mapping table $P_M = (363, 109)$
- Let us choose $k = 2$

$$kP_B = (166, 219)$$

$$C_m = [(483, 112), (271, 49)]$$

3.3 Integration of PUF and ECC

The integration of a Physically Unclonable Function (PUF) and Elliptic Curve Cryptography (ECC) combines the unique hardware-based security features of PUFs with the robust cryptographic security of ECC. This integration ensures the secure generation, encryption, and transmission of PUF responses.

Steps in the Integration Process:

3.3.1 Challenge-Response Pair Generation:

- A challenge is sent to the PUF.
- The PUF processes the challenge and generates a unique binary response based on its inherent physical characteristics.

3.3.2 Response Encryption Using ECC:

The unique PUF response is encrypted using ECC to protect it from machine learning attacks, interception and unauthorized access.

i. Key Management:

- **Public Key Distribution:** The public key, which is used for encryption, is distributed to the entities that will encrypt the PUF responses.
- **Private Key Storage:** The private key, which is used for decryption, is securely stored by the entity that needs to decrypt the PUF responses.

ii. Encryption Process:

- The PUF response is encrypted using the recipient's public key. This involves generating a shared secret using ECC and then using this secret to encrypt the PUF response.

3.4. Machine Learning Attacks

In the case of PUFs the term ML attack is used equivalently to ML algorithm. ML attacks are used to learn the response behaviour of PUFs. To train the model by a ML attack CRPs, which are already known, are used. For challenges their responses are not known the model then can make predictions about whose responses.

To categorize ML techniques by their approach the following three different learning approach classes are used:

- **supervised learning:** The data includes input values and their desired output values for the ML algorithm. In the training process the ML algorithm has to make predictions about the data and in this way trains its model. The goal is to learn the rules how the input values mapped onto the output values.

- **unsupervised learning:** The data includes input values only. These data are used in the training process, so that the ML algorithm can find structures in the data. The goal can be to reveal that there is a structure in the data or to learn the structure. If the algorithm learned the structure, it will be able to repeat it to a certain level.
- **semi-supervised learning:** The data includes input values with their desired output values and input values without their output values. In the training process the ML algorithm has to structure the data and make already prediction for the data without output values.
- **reinforcement learning:** The ML algorithm is trained by reward and punishment to maximize its performance in a dynamic environment. This learning approach is based on a common way humans learn. The algorithm can choose an action to change the state of the environment over to another state. Depending on this transition the algorithm gets rewarded. The goal of the algorithm is to collect as much reward as possible.

Different ML attacks are used to attack PUFs. First, the required inputs for the ML attacks and the problem they are solving are explained. After that, each ML technique is outlined by its own.

The ML attack creates a PUF model of the attacked PUF, which acts as binary classifier. A binary classifier divides the samples, in the case of PUFs these are challenges, of a given set into two different classes 0 or 1 similar to the responses of a PUF. To build this model a training set $D = \{(z_1, e_1), \dots, (z_l, e_l)\}$ is necessary, whereas the integer l is the number of training set samples $d_i = (z_i, e_i)$ that are already classified by the attacked PUF. Let l be the number of input values of a sample d_i . This sample is then defined by its values $z_i = (z_{i,1}, \dots, z_{i,l})$ and is classified into the class e_i . With that training set the model can be trained by the ML algorithm. To evaluate the built model, a test set is created the same way and used to compare the output of the PUF model and the attacked PUF.

The most PUFs, known by the year 2014, which can be built by an electrical circuit, are related to internal values of the circuit. These values determine the response for a given challenge. The goal of the ML attacks is to find a representation of these values, so that the created model classifies samples in the same way as the attacked PUF. How many samples of the test set the model classifies the same way as the PUF defines how precise the model has been trained. Accordingly, the values learned by the ML attack do not have to be exact the same values that defined the attacked PUF to classify samples of a set the same way as the PUF.

Whether a ML attack can find a binary classifier for a given set depends on the attack and if the set of samples is linear separable. Instead of a set of samples this also applies to binary functions

$$f : \mathbb{R}^n \rightarrow \{0, 1\}.$$

One example for a not linear separable function is the XOR function. Some ML attacks, e.g. single-layer perceptron, can only learn linear separable sets and can therefore not learn XOR functions. Two subsets $A \subseteq \mathbb{R}^n$, $B \subseteq \mathbb{R}^n$ are called linear separable if all their vectors $\vec{a} = (a_1, \dots, a_n) \in A$, $\vec{b} = (b_1, \dots, b_n) \in B$ can be separated by

$$\sum_{i=1}^n w_i a_i \leq w_{n+1} < \sum_{j=1}^n w_j b_j$$

whereas $w_1, \dots, w_{n+1}$ are $n + 1$ real numbers. The separating hyperplane is defined by the vector $\vec{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$ that satisfies $\sum_{i=1}^n w_i x_i = w_{n+1}$. In the case of building a classifier by the ML attacks the two subset A and B are subsets of the given set used to train the model.

The aim of attacks on Arbiter PUF is to predict responses for challenges their responses are unknown. To accomplish this on the one hand it is possible to reveal the intrinsic secret represented by the inaccuracies of the Arbiter PUF circuit. On the other hand, the challenge response behaviour of the PUF can be adapted. Both methods can then be used to impersonate the attacked PUF.

Since Arbiter PUFs were one of the most promising PUF implementation, multiple attacks have been developed. One of the first attacks reveals the internal values of the Arbiter PUF by linear programming. This approach solves a linear equation system after collecting enough CRPs, because an Arbiter PUF can be represented as linear threshold function. For a good result the stability of the responses of the CRPs should be as high as possible. As the response of every CRP includes some noise, one strategy of the attacker is to reduce the noise by using majority vote, i.e. evaluating the same challenge multiple times and picking the most appeared response. Therefore, the attacker has to be able to evaluate chosen challenges.

The linear programming attack is a non-invasive modeling attack where the attacker needs to collect CRPs to build a model of the PUF for example by a ML algorithm. Subsequently, the model can predict responses for new challenges, which have not been used to create the model itself. How precise these predictions are depends on the number of captured CRPs and used parameters to build the model. The result and the model building performance vary for different ML techniques. How the attacker gets to know the CRPs in a real use case depends on the way the PUF is used and the CRPs transmitted and is therefore not part of this work.

Arbiter PUFs are strong PUFs and their operation frequency is relatively high. For that reason they provide the option to freely access a large amount of CRPs. On the contrary, some PUFs post-process their response in a protected chip environment to make the PUF response itself inaccessible to protect the PUF.

3.4.1 Logistic Regression

Logistic Regression is a widely used machine learning algorithm for binary classification problems. It has been explored as a technique for launching Machine Learning (ML) attacks on Physical Unclonable Functions (PUFs) to model their challenge-response behavior and predict responses for unseen challenges.

Logistic Regression is particularly suitable for attacking PUFs because it can effectively capture the relationship between the challenges (input features) and the binary responses generated by the PUF. By modeling this relationship, the attacker can predict the likelihood of obtaining a '0' or '1' response for a given challenge.

In the context of ML attacks on PUFs, the Logistic Regression algorithm can be applied to model the challenge-response behavior of a PUF using a set of known Challenge-Response Pairs (CRPs). The attacker's goal is to train a Logistic Regression model on a subset of CRPs and then use the trained model to predict responses for new, unseen challenges.

The attack proceeds as follows:

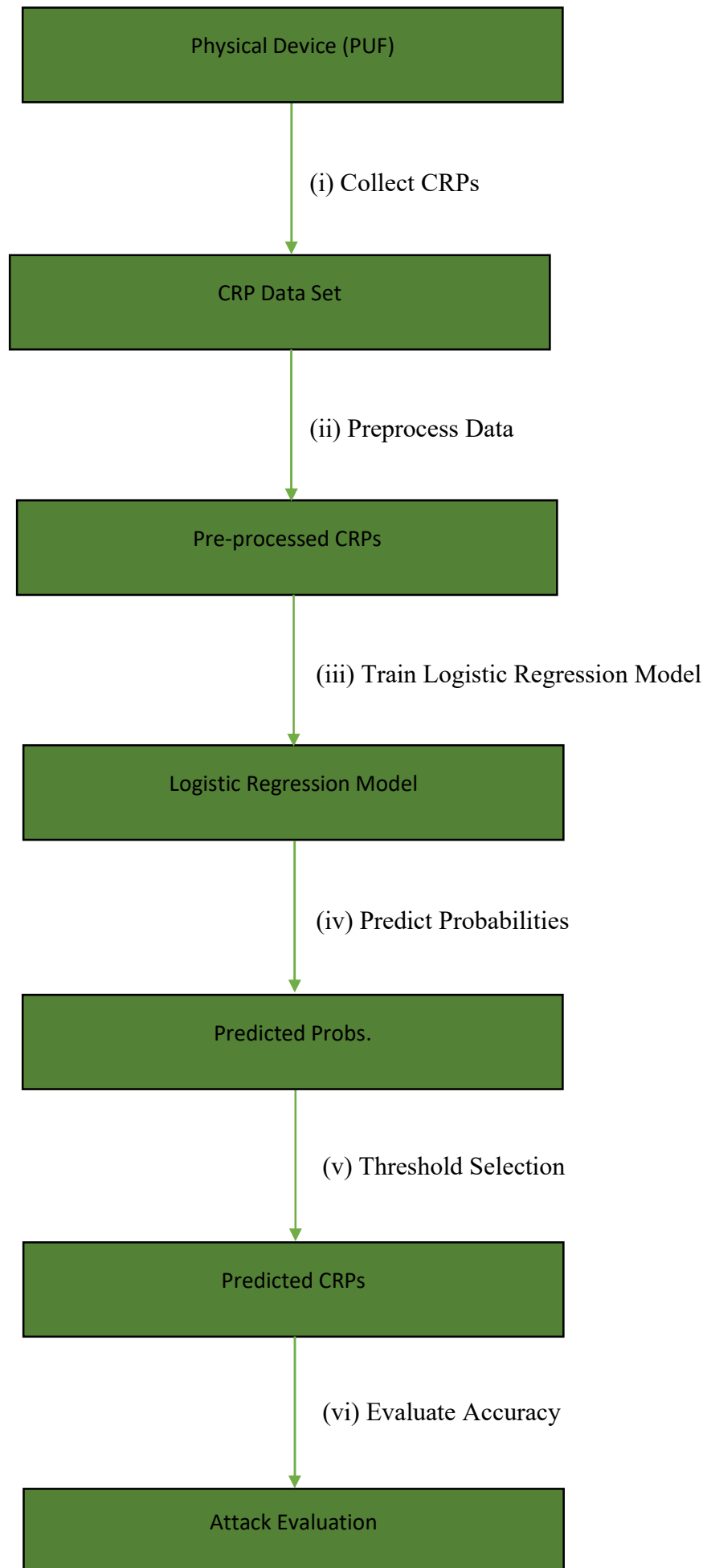
- i. Data Collection: The attacker first obtains a set of CRPs from the target PUF by applying various challenges and recording the corresponding responses.
- ii. Data Preprocessing: The collected CRPs are pre-processed and formatted to serve as input for the Logistic Regression model. This typically involves encoding the challenges as numerical features and the responses as binary labels (e.g., 0 or 1).
- iii. Model Training: A Logistic Regression model is trained using a portion of the collected CRPs. The model learns the relationship between the challenges and the corresponding responses by estimating the coefficients of the logistic function.

iv. Prediction: Once the Logistic Regression model is trained, it can be used to predict responses for new, unseen challenges. The attacker provides the unseen challenges as input to the trained model, and the model outputs the predicted probabilities of obtaining a '0' or '1' response.

v. Threshold Selection: The attacker selects a threshold value (e.g., 0.5) to convert the predicted probabilities into binary predictions. Challenges with predicted probabilities above the threshold are classified as '1', and those below the threshold are classified as '0'.

vi. Evaluation: The attacker evaluates the accuracy of the Logistic Regression model's predictions by comparing them with the actual responses from the PUF. If the prediction accuracy is sufficiently high, the attacker can effectively model the PUF's behavior and potentially break its security.

The effectiveness of the Logistic Regression attack on PUFs depends on several factors, including the number and diversity of CRPs available for training, the complexity of the PUF design, and the presence of any countermeasures or obfuscation techniques employed to protect the PUF.



Flow Chart of Logistic Regression

The diagram illustrates the general workflow of a Logistic Regression attack on PUFs. It begins with collecting CRPs from the target PUF (1), followed by data preprocessing (2) to prepare the CRPs for training. The preprocessed CRPs are then used to train a Logistic Regression model (3), which learns the relationship between challenges and responses. Once trained, the model can predict probabilities for new, unseen challenges (4). The attacker selects a threshold value (5) to convert these probabilities into binary predictions (CRPs). Finally, the attacker evaluates the accuracy of the predicted CRPs (6) to assess the effectiveness of the attack.

Logistic Regression attacks have been explored in the literature as a straightforward and effective approach for modeling PUFs with binary responses. However, like other ML attacks, they can be mitigated by employing countermeasures such as introducing non-linearity, obfuscation techniques, lightweight PUF designs, and cryptographic primitives like Elliptic Curve Cryptography (ECC) to enhance the security of PUFs against ML attacks.

3.4.2 Random Forest

Random Forest is a powerful ensemble learning technique that combines multiple decision trees to improve prediction accuracy and robustness. In the context of Machine Learning (ML) attacks on Physical Unclonable Functions (PUFs), Random Forest has emerged as an effective approach for modeling the behavior of PUFs and predicting responses for unseen challenges.

The Random Forest algorithm operates by constructing multiple decision trees during the training phase. Each decision tree is built using a random subset of the training data and a random subset of features (challenges in the case of PUFs). This randomization process helps to introduce diversity among the individual decision trees, reducing the risk of overfitting and improving the overall generalization ability of the model.

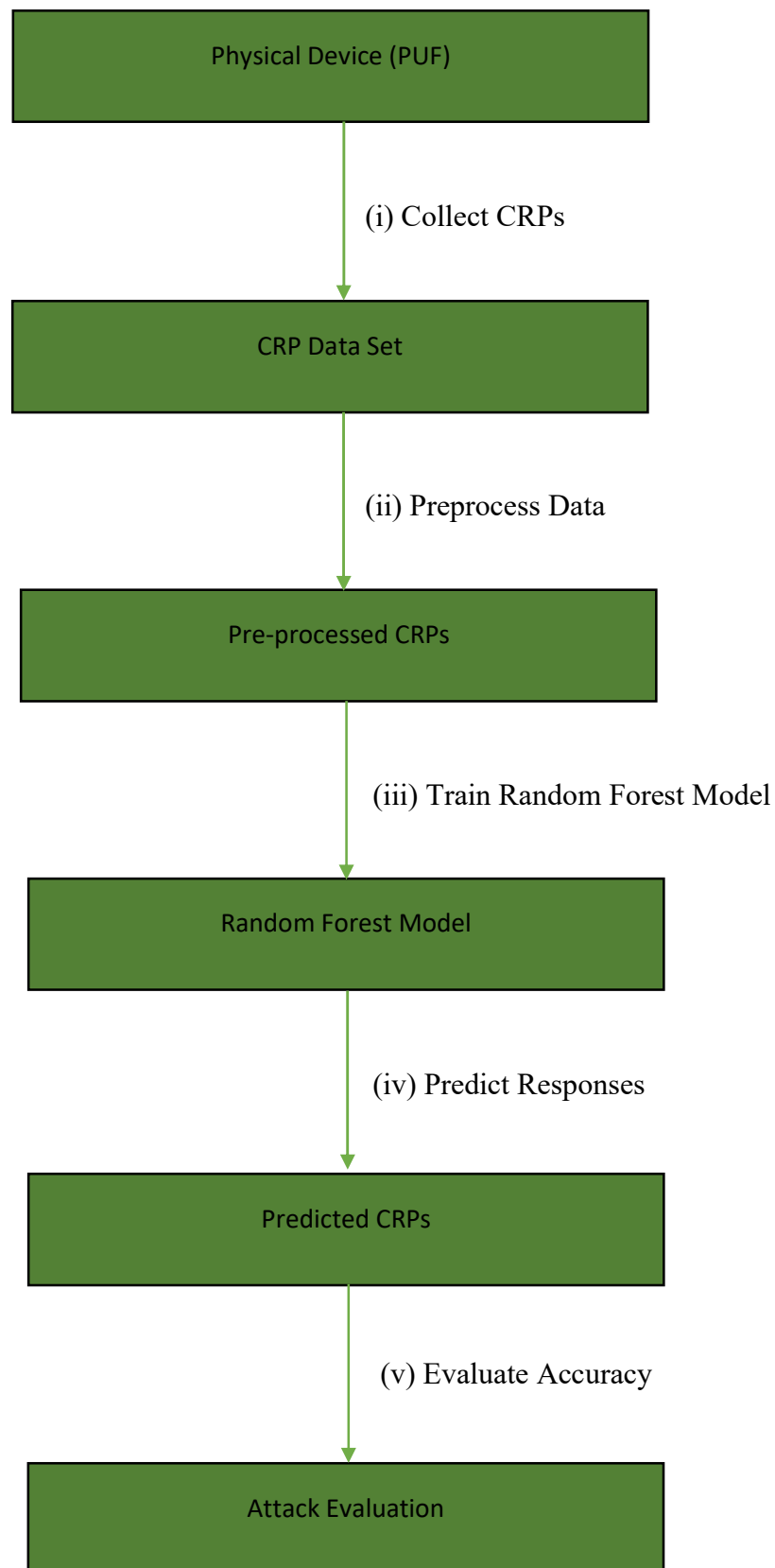
During the prediction phase, the Random Forest model aggregates the predictions from all the individual decision trees. The final prediction is typically determined by a majority vote or an average of the individual tree predictions. This ensemble approach helps to mitigate the potential biases or errors of individual decision trees, leading to more accurate and reliable predictions.

In the context of ML attacks on PUFs, the Random Forest algorithm can be applied to model the challenge-response behavior of a PUF using a set of known Challenge-Response Pairs (CRPs). The attacker's goal is to train a Random Forest model on a subset of CRPs and then use the trained model to predict responses for new, unseen challenges.

The attack proceeds as follows:

- i. Data Collection: The attacker first obtains a set of CRPs from the target PUF by applying various challenges and recording the corresponding responses.
- ii. Data Preprocessing: The collected CRPs are preprocessed and formatted to serve as input for the Random Forest model. This may involve encoding the challenges and responses into suitable numerical representations.
- iii. Training: A Random Forest model is trained using a portion of the collected CRPs. During the training process, multiple decision trees are constructed, each based on a random subset of the training data and a random subset of features (challenges).
- iv. Prediction: Once the Random Forest model is trained, it can be used to predict responses for new, unseen challenges. The attacker provides the unseen challenges as input to the trained model, and the model outputs the predicted responses based on the collective decisions of the individual decision trees.
- v. Evaluation: The attacker evaluates the accuracy of the Random Forest model's predictions by comparing them with the actual responses from the PUF. If the prediction accuracy is sufficiently high, the attacker can effectively model the PUF's behavior and potentially break its security.

The effectiveness of the Random Forest attack on PUFs depends on several factors, including the number and diversity of CRPs available for training, the complexity of the PUF design, and the presence of any countermeasures or obfuscation techniques employed to protect the PUF.



Flow Chart of Random Forest

The diagram illustrates the general workflow of a Random Forest attack on PUFs. It begins with collecting CRPs from the target PUF (1), followed by data preprocessing (2) to prepare the CRPs for training. The preprocessed CRPs are then used to train a Random Forest model (3), which consists of multiple decision trees. Once trained, the model can predict responses for new, unseen challenges (4). Finally, the attacker evaluates the accuracy of the predicted CRPs (5) to assess the effectiveness of the attack.

Random Forest attacks on PUFs have been studied extensively in the literature, and various countermeasures have been proposed to mitigate their effectiveness. These countermeasures include introducing non-linearity, obfuscation techniques, lightweight PUF designs, and cryptographic primitives like Elliptic Curve Cryptography (ECC) to enhance the security of PUFs against ML attacks.

3.4.3 Kernel Ridge Regression

Kernel Ridge Regression is a kernel-based machine learning algorithm that combines the principles of ridge regression and the kernel trick. It has been explored as an effective technique for modeling the behavior of Physical Unclonable Functions (PUFs) and launching Machine Learning (ML) attacks to predict responses for unseen challenges.

The kernel trick in Kernel Ridge Regression allows the algorithm to operate in a high-dimensional feature space, which can capture complex non-linear relationships between challenges and responses. This makes it particularly suitable for attacking PUFs, which often exhibit non-linear behavior due to the intricate physics and manufacturing variations involved in their construction.

In the context of ML attacks on PUFs, the Kernel Ridge Regression algorithm can be applied to model the challenge-response behavior of a PUF using a set of known Challenge-Response Pairs (CRPs). The attacker's goal is to train a Kernel Ridge Regression model on a subset of CRPs and then use the trained model to predict responses for new, unseen challenges.

The attack proceeds as follows:

- i. Data Collection: The attacker first obtains a set of CRPs from the target PUF by applying various challenges and recording the corresponding responses.

ii. Data Preprocessing: The collected CRPs are preprocessed and formatted to serve as input for the Kernel Ridge Regression model. This may involve encoding the challenges and responses into suitable numerical representations.

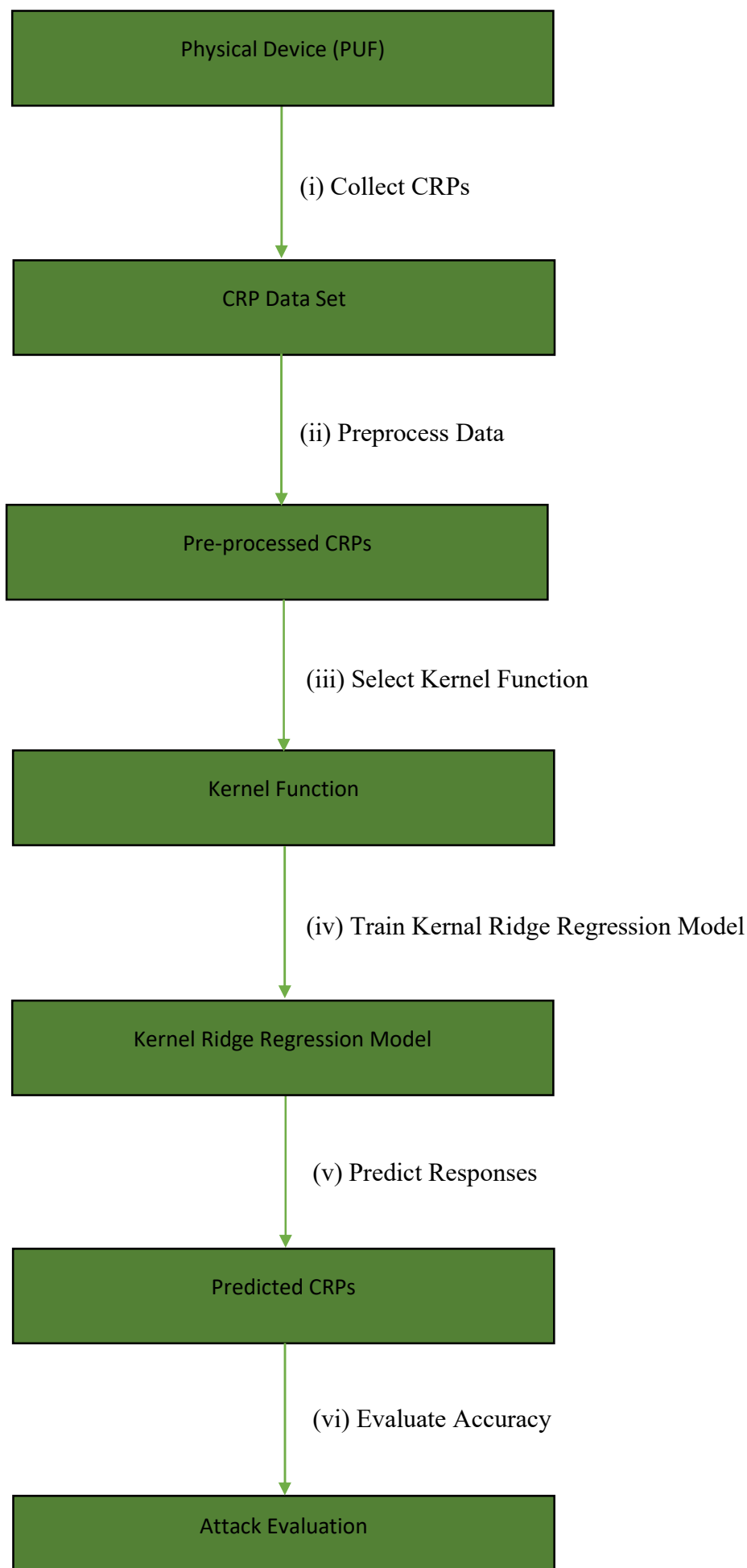
iii. Kernel Function Selection: The attacker selects an appropriate kernel function (e.g., Gaussian kernel, polynomial kernel) that can effectively capture the non-linear relationships between challenges and responses in the PUF.

iv. Model Training: A Kernel Ridge Regression model is trained using a portion of the collected CRPs. The kernel function is used to map the input data into a higher-dimensional feature space, where ridge regression is performed to find the optimal model parameters.

v. Prediction: Once the Kernel Ridge Regression model is trained, it can be used to predict responses for new, unseen challenges. The attacker provides the unseen challenges as input to the trained model, and the model outputs the predicted responses based on the learned non-linear mapping and regression coefficients.

vi. Evaluation: The attacker evaluates the accuracy of the Kernel Ridge Regression model's predictions by comparing them with the actual responses from the PUF. If the prediction accuracy is sufficiently high, the attacker can effectively model the PUF's behavior and potentially break its security.

The effectiveness of the Kernel Ridge Regression attack on PUFs depends on several factors, including the number and diversity of CRPs available for training, the complexity of the PUF design, the choice of an appropriate kernel function, and the presence of any countermeasures or obfuscation techniques employed to protect the PUF.



Flow Chart of Kernel Ridge Regression

The diagram illustrates the general workflow of a Kernel Ridge Regression attack on PUFs. It begins with collecting CRPs from the target PUF (1), followed by data preprocessing (2) to prepare the CRPs for training. An appropriate kernel function is selected (3) to capture the non-linear relationships between challenges and responses. The preprocessed CRPs and the chosen kernel function are then used to train a Kernel Ridge Regression model (4). Once trained, the model can predict responses for new, unseen challenges (5). Finally, the attacker evaluates the accuracy of the predicted CRPs (6) to assess the effectiveness of the attack.

Kernel Ridge Regression attacks have been explored in the literature as an effective approach for modeling PUFs with non-linear behavior. However, like other ML attacks, they can be mitigated by employing countermeasures such as introducing non-linearity, obfuscation techniques, lightweight PUF designs, and cryptographic primitives like Elliptic Curve Cryptography (ECC) to enhance the security of PUFs against ML attacks.

3.4.4 Gradient Boosting

Gaussian Gradient Boosting is a powerful machine learning technique that combines the principles of gradient boosting and Gaussian processes. It has been explored as an effective approach for modeling the behavior of Physical Unclonable Functions (PUFs) and launching Machine Learning (ML) attacks to predict responses for unseen challenges.

Gaussian Gradient Boosting is particularly well-suited for attacking PUFs because it can effectively capture complex non-linear relationships between challenges and responses. Additionally, it provides an estimate of the uncertainty associated with its predictions, which can be useful in assessing the confidence of the attack.

In the context of ML attacks on PUFs, the Gaussian Gradient Boosting algorithm can be applied to model the challenge-response behavior of a PUF using a set of known Challenge-Response Pairs (CRPs). The attacker's goal is to train a Gaussian Gradient Boosting model on a subset of CRPs and then use the trained model to predict responses for new, unseen challenges.

The attack proceeds as follows:

- i. **Data Collection:** The attacker first obtains a set of CRPs from the target PUF by applying various challenges and recording the corresponding responses.

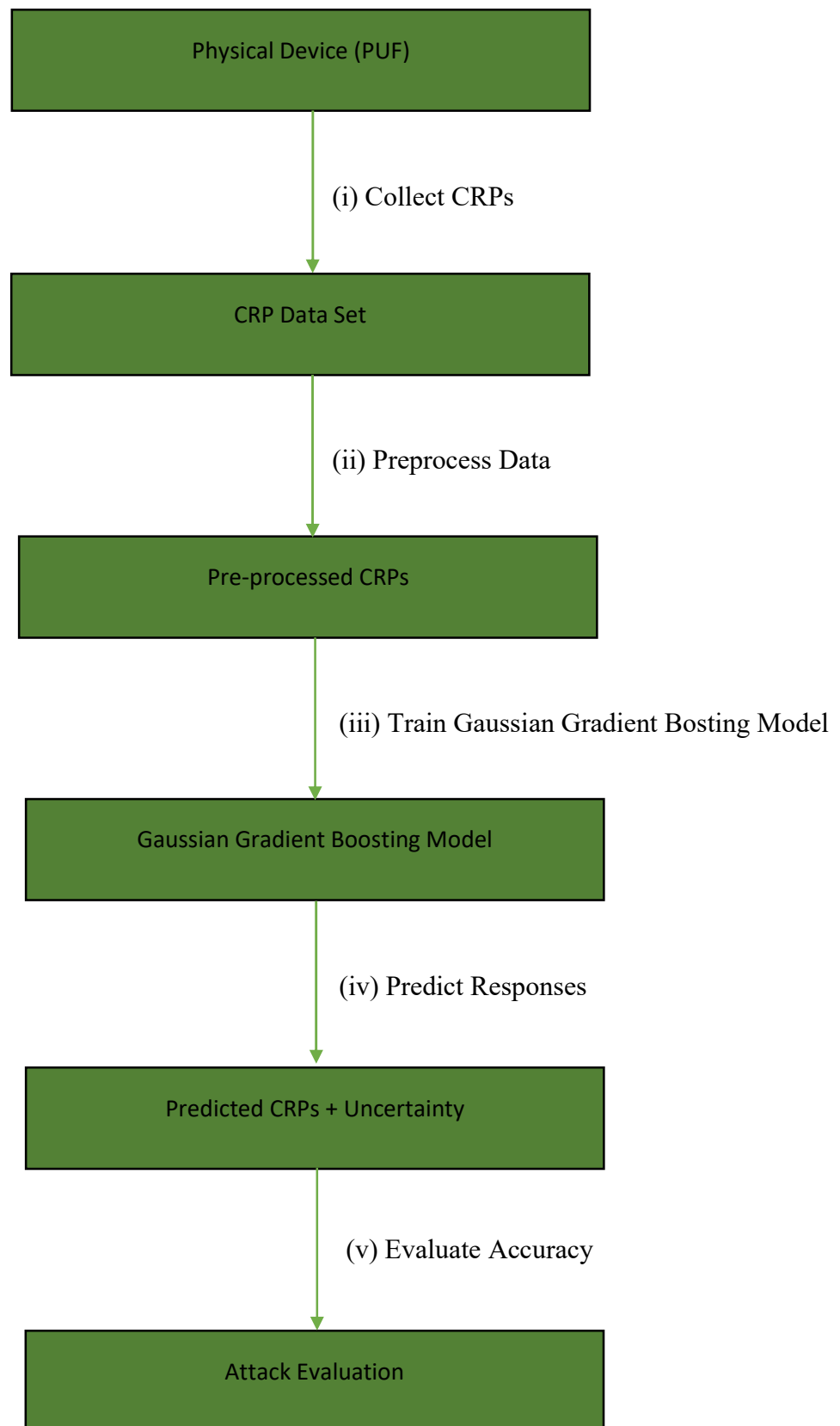
ii. Data Preprocessing: The collected CRPs are preprocessed and formatted to serve as input for the Gaussian Gradient Boosting model. This may involve encoding the challenges and responses into suitable numerical representations.

iii. Model Training: A Gaussian Gradient Boosting model is trained using a portion of the collected CRPs. The model iteratively combines weak learners (typically decision trees) in a stage-wise fashion, with each iteration focused on minimizing the residual errors from the previous iteration.

iv. Prediction: Once the Gaussian Gradient Boosting model is trained, it can be used to predict responses for new, unseen challenges. The attacker provides the unseen challenges as input to the trained model, and the model outputs the predicted responses along with an estimate of the prediction uncertainty.

v. Evaluation: The attacker evaluates the accuracy of the Gaussian Gradient Boosting model's predictions by comparing them with the actual responses from the PUF. If the prediction accuracy is sufficiently high, the attacker can effectively model the PUF's behavior and potentially break its security. The prediction uncertainty estimates can also be used to assess the confidence in the attack.

The effectiveness of the Gaussian Gradient Boosting attack on PUFs depends on several factors, including the number and diversity of CRPs available for training, the complexity of the PUF design, the choice of appropriate hyperparameters for the model, and the presence of any countermeasures or obfuscation techniques employed to protect the PUF.



Flow Chart of Gradient Boosting

The diagram illustrates the general workflow of a Gaussian Gradient Boosting attack on PUFs. It begins with collecting CRPs from the target PUF (1), followed by data preprocessing (2) to prepare the CRPs for

training. The pre-processed CRPs are then used to train a Gaussian Gradient Boosting model (3), which iteratively combines weak learners to capture complex non-linear relationships. Once trained, the model can predict responses for new, unseen challenges (4), along with an estimate of the prediction uncertainty. Finally, the attacker evaluates the accuracy of the predicted CRPs (5) to assess the effectiveness of the attack, potentially utilizing the prediction uncertainty estimates as well.

Gaussian Gradient Boosting attacks have been explored in the literature as an effective approach for modeling PUFs with non-linear behavior and estimating prediction uncertainties. However, like other ML attacks, they can be mitigated by employing countermeasures such as introducing non-linearity, obfuscation techniques, lightweight PUF designs, and cryptographic primitives like Elliptic Curve Cryptography (ECC) to enhance the security of PUFs against ML attacks.

3.5 Evaluation and Testing

3.5.1 Security Testing

The security of the system was tested against various ML attacks:

- **Training ML Models:** Different ML models (e.g., linear regression, neural networks) were trained using CRPs.
- **Evaluating Predictability:** The predictability of encrypted responses was analyzed to assess the effectiveness of ECC in obfuscating the relationship between challenges and responses.

5.5.2 Comparative Analysis

The system was compared to traditional PUF implementations without ECC encryption to highlight the improvements in security and resistance to ML attacks.

This methodology provides a comprehensive approach to designing, implementing, and testing an ECC-encrypted PUF system. The detailed steps ensure the system's robustness and security, making it a viable solution for enhancing hardware security against ML attacks.

CHAPTER 4

IMPLEMENTATION

WE have implemented a security system integrating Physically Unclonable Functions (PUFs) with Elliptic Curve Cryptography (ECC) in Python. The steps include implementing the PUF, generating ECC keys, integrating PUF responses with ECC encryption, and evaluating the security and performance of the system. This section provides the theoretical background and outlines the Python-specific considerations for each step.

4.1. Implementing the PUF

4.1.1 Arbiter PUF Design:

- **Setup:** The arbiter PUF consists of a series of delay stages, each made up of a pair of multiplexers (MUX) and an arbiter at the end.
- **Delay Stages:** Each stage has two multiplexers that direct the signal along different paths. Manufacturing process differences introduce slight variations in the delay of each path.
- **Arbiter:** After passing through the delay stages, the signal reaches an arbiter that determines which path's signal arrived first, producing a binary response (0 or 1).

4.1.2 Challenge-Response Mechanism:

- **Challenges:** The input to the PUF is a challenge, a binary string that controls the multiplexers at each stage.
- **Responses:** The output is a response, a binary string generated based on the delay variations influenced by the challenge. Each challenge-response pair (CRP) is unique due to the intrinsic properties of the PUF.

4.1.3 Python Considerations:

- Simulation: In Python, to simulate the arbiter PUF we used lists to represent the delay stages and logical operations to simulate the delays and arbitration.
- Libraries: Used library Pandas for efficient bit manipulation.

4.2. ECC Key Generation

Elliptic Curve Cryptography (ECC) provides robust encryption with smaller key sizes and enhanced security compared to other cryptographic methods.

4.2.1 Key Generation Process:

- Elliptic Curve Equation: The general form of an elliptic curve equation is $y^2 = x^3 + ax + b$ over a finite field.
- Point Multiplication: The public key Q is obtained by multiplying the base point G by the private key.

4.2.2 Python Considerations:

- Libraries: Use libraries such as NumPy and pandas for ECC key generation and cryptographic operations.

4.3 Integrating PUF Responses with ECC Encryption

To enhance security, the responses generated by the PUF are encrypted using ECC before being used in any security protocol or stored.

4.3.1 Encryption Process:

- Plaintext: The plaintext is the PUF response generated for a given challenge.
- Public Key: Use the receiver's public key for encryption. The encryption process involves converting the plaintext into a point on the elliptic curve.
- Ciphertext: The encrypted PUF response is represented as a point on the elliptic curve, ensuring that it can only be decrypted by the corresponding private key.

4.3.2 Python Considerations:

- Libraries: We use libraries such as NumPy and pandas for ECC key generation and cryptographic operations.

4.4 Evaluation and Testing

After integrating the PUF and ECC, it is essential to evaluate the system's security and performance to ensure it meets the desired criteria.

4.4.1 Security Evaluation:

- Machine Learning Resistance: We assessed the system's ability to resist machine learning attacks by testing with various models and datasets.

4.4.2 Python Considerations:

- Machine Learning Resistance: Use machine learning libraries like sklearn, train_test_split, StandardScaler, KernelRidge, GradientBoostingRegressor, r2_score, MinMaxScaler, RandomForestRegressor, etc to test and evaluate the system's resistance to machine learning attacks.

Implementing a security system that integrates PUFs with ECC involves designing a reliable PUF, generating secure ECC keys, and combining these technologies to create a robust defense against machine learning and other cryptographic attacks. By encrypting PUF responses, the system ensures that even if the responses are intercepted, they cannot be used to compromise the security of the device. The implementation process, from designing the PUF to integrating it with ECC and evaluating the system's performance, highlights the feasibility and effectiveness of this approach in enhancing hardware security. Using Python, with its rich set of libraries and tools, facilitates efficient and secure implementation of the system.

CHAPTER 5

RESULTS AND DISCUSSION

The results and discussion section provides an in-depth analysis of the performance and security evaluation of the implemented system integrating Physically Unclonable Functions (PUFs) with Elliptic Curve Cryptography (ECC). This section includes a detailed examination of the system's resilience against machine learning attacks. The results are interpreted to understand the effectiveness and practicality of the integrated system.

5.1 Machine Learning Resistance:

To evaluate the resistance of the R-PUF system to machine learning attacks, various machine learning models were trained on the challenge-response pairs (CRPs) generated by the PUF. The goal was to see if these models could predict the PUF responses based on the challenges.

5.1.1 Methodology:

- A dataset of CRPs was generated by simulating the arbiter PUF.
- Machine learning models, like Kernel Ridge Regression, Random Forest Regression, Gradient Boosting and Logistic regression were trained on a subset of this dataset.
- The models' performance was measured using metrics such as training score and testing score.
- Then a dataset of CRPs was generated by R- PUF and machine learning models were trained on a subset of these dataset.
- The models' performance was measured using metrics such as training score and testing score.
- The two models' performance is then compared.

5.1.2 Results:

- Kernel Ridge Regression: The model showed an accuracy of 0.084%, showing limited predictive power.
- Random Forest Regression: The model showed an accuracy of 46.96%
- Gradient Boosting: The model showed an accuracy of 0.381%

- Logistic Regression: The model showed an accuracy of around 54%, indicating it could not effectively learn the pattern in the CRPs.

```
# Make predictions
y_pred_train = krr_model.predict(X_train)
y_pred_test = krr_model.predict(X_test)

# Evaluate the model
training_score = krr_model.score(X_train, y_train)
testing_score = krr_model.score(X_test, y_test)

print("Training Score:", training_score)
print("Testing Score:", testing_score)
```

Training Score: 0.00024989915076489844
Testing Score: 0.0008411645136199297

Fig 5.2.1a Kernel Ridge Regression

```
# Make predictions
y_pred_train = gbm_model.predict(X_train)
y_pred_test = gbm_model.predict(X_test)

# Evaluate the model using R-squared
training_r2 = abs(r2_score(y_train, y_pred_train))
testing_r2 = abs(r2_score(y_test, y_pred_test))

print("Training R-squared:", training_r2)
print("Testing R-squared:", testing_r2)
```

Training R-squared: 0.03377512311917741
Testing R-squared: 0.0038174112774496116

Fig 5.2.1b Gradient Boosting Regression

```
11877    14584579297234455733
5421     14584579297234455733
3855     14584579297234455733
4414     14584579297234455733
Name: Encrypted Response, Length: 2400, dtype: uint64
Training Score: 0.7965054712121712
Testing Score: 0.46964988710098043

[2]: from sklearn.ensemble import GradientBoostingRegressor
      from sklearn.model_selection import train_test_split
      from sklearn.metrics import r2_score
```

Fig 5.21.c Random Forest Regression

```
# Initialize and fit the Logistic regression model
model = LogisticRegression(max_iter=12000)
model.fit(X_train, y_train)

y_pred_train = model.predict(X_train).reshape(-1, 1)
y_pred_test = model.predict(X_test).reshape(-1, 1)

# Evaluate the model
training_score = model.score(y_pred_train, y_train)
testing_score = model.score(y_pred_test, y_test)

print("Training Accuracy:", training_score)
print("Testing Accuracy:", testing_score)
```

Training Accuracy: 0.5408333333333334
Testing Accuracy: 0.5425

Fig 5.2.1d Logistic Regression

The following table shows the comparison of results between the Arbiter PUF and R-PUF for all the Machine Learning models used.

ML ATTACK	Arbiter PUF		R-PUF	
	Training Score	Testing Score	Training score	Testing score
Kernel Ridge Regression	0.640125526	0.639114099	0.000249899	0.000841164
Random Forest	0.928724446	0.493983948	0.796505471	0.4696498871
Gradient Boosting	0.042792130	0.046371321	0.033775123	0.003817411
Logistic Regression	0.606909090	0.61	0.54083333	0.5425

Table 5.2 Comparison of ML attacks between the Arbiter PUF and R-PUF

5.2 Discussion:

The low accuracy of the machine learning models demonstrates the effectiveness of the R-PUF system in thwarting machine learning attacks. The encryption of PUF responses using ECC adds an additional layer of security by obfuscating the direct relationship between challenges and responses, making it difficult for machine learning models to learn any meaningful patterns. As R-PUF is encrypted, the attacker cannot directly obtain the correspondence between the challenge and response. The attacker can only model the R-PUF if he knows the key.

CHAPTER 6

Conclusion and Future work

6.1 Conclusion

Encrypting PUF responses using Elliptic Curve Cryptography (ECC) effectively mitigates the risk of machine learning attacks and significantly enhances the overall security of PUF-based systems. By combining the inherent physical security provided by PUFs, which generate unique and unpredictable responses due to manufacturing variations, with the robust cryptographic protection of ECC, a highly secure and efficient security mechanism is achieved. This integration obfuscates the relationship between challenges and responses, making it exceedingly difficult for attackers to use machine learning models to predict responses.

6.1.1 Benefits of R-PUF

i. Thwarting Machine Learning Attacks:

Machine learning models rely on patterns in data to make predictions. By encrypting PUF responses with ECC, the direct relationship between challenges and responses is hidden, preventing attackers from gathering the necessary data to train their models effectively. This obfuscation disrupts the learning process, rendering machine learning attacks ineffective.

ii. Enhanced Security:

ECC provides strong cryptographic security with relatively small key sizes, making it efficient and practical for a wide range of applications. The use of ECC ensures that even if an attacker intercepts the encrypted PUF responses, they cannot decrypt the data without access to the private key. This robust encryption method protects against unauthorized access and tampering.

iii. Efficiency and Scalability:

ECC's efficiency with smaller key sizes and lower computational requirements makes it ideal for use in resource-constrained environments such as IoT devices and embedded systems. This efficiency allows for the widespread adoption of PUF-ECC integration in various applications, providing scalable security solutions without compromising performance.

iv. Robustness:

The physical security of PUFs, derived from their unique manufacturing variations, combined with the mathematical security of ECC, creates a comprehensive and resilient security solution. This layered defense mechanism ensures that even if one security layer is compromised, the overall system remains secure, providing robust protection against a variety of attack vectors.

In conclusion, the integration of PUFs and ECC creates a highly secure and efficient system suitable for a wide range of applications, including secure authentication, encryption, and data protection in hardware-constrained environments. This approach not only enhances security but also ensures the integrity and confidentiality of critical information, providing a robust defense against various attack vectors. The ongoing research and development in this field will continue to drive innovation and improve the effectiveness of hardware-based security solutions, ensuring that PUF-ECC integration remains at the forefront of secure technology advancements.

6.2 Future work

The integration of Physically Unclonable Functions (PUFs) with Elliptic Curve Cryptography (ECC) represents a significant advancement in hardware security, yet it also opens the door to numerous possibilities for further exploration and enhancement. Future research and development in this field can focus on several key areas to maximize the potential of PUF-ECC integration. These areas include advanced PUF designs, optimization of ECC algorithms, improved machine learning resistance, applications in IoT and cyber-physical systems, and efforts toward standardization and interoperability.

i. Exploration of New PUF Architectures:

Optical PUFs: Optical PUFs leverage light-based physical characteristics to generate unique responses. These PUFs can offer high entropy and resistance to traditional electronic attack vectors. However, their practical implementation, cost-effectiveness, and integration into existing systems require extensive research. Future work could focus on optimizing the design and fabrication processes to make optical PUFs more viable for commercial applications.

Memristor-Based PUFs: Memristor-based PUFs use memory resistors, which exhibit varying resistance states due to their physical properties. These PUFs can provide enhanced security and efficiency, making

them suitable for embedded systems and IoT devices. Future research could delve into improving the stability and reliability of memristor-based PUFs, exploring their response time, integration challenges, and potential applications in various security domains.

ii. Enhanced Reliability and Robustness:

Environmental Stability: Ensuring that PUF responses remain consistent across different environmental conditions is crucial for their reliability. Future research can focus on developing techniques to enhance PUF stability despite temperature fluctuations, humidity changes, and device aging. This might involve the use of error correction codes, compensation algorithms, and other methods to maintain the reliability of PUFs under diverse conditions.

Self-Healing PUFs: Investigating self-healing PUFs that can adapt and recalibrate their responses over time to maintain stability and reliability. This capability could involve developing adaptive algorithms that monitor and adjust PUF responses to compensate for physical degradation and environmental variations, ensuring consistent performance throughout the device's lifespan.

iii. Optimization of ECC Algorithms

Resource-Constrained Devices: Researching and developing more efficient ECC algorithms tailored for low-power and resource-constrained devices, such as IoT sensors and wearable technology. This involves optimizing computational efficiency and reducing memory usage while maintaining strong security properties, making ECC more viable for widespread use in these environments.

Hardware Acceleration: Utilizing hardware accelerators, such as dedicated cryptographic co-processors or FPGA-based implementations, to speed up ECC operations and improve overall system performance. Hardware acceleration can significantly enhance the practicality of ECC encryption for real-time applications, particularly in devices with limited processing power.

The integration of PUFs and ECC opens numerous opportunities for further research and development. Future work can focus on exploring advanced PUF designs, optimizing ECC algorithms for resource-

constrained environments, enhancing resistance to machine learning attacks, and applying this technology to IoT and cyber-physical systems. Additionally, contributing to standardization efforts and ensuring interoperability will help establish a broad base of secure applications and facilitate global adoption.

Research into new PUF architectures, such as optical and memristor-based PUFs, promises to further improve security and reliability. The development of more efficient ECC implementations and post-quantum cryptographic solutions will ensure that PUF-ECC systems remain robust against evolving threats. Furthermore, applying PUF-ECC integration in IoT devices and secure supply chain management can significantly enhance security in various critical applications. This approach not only enhances security but also ensures the integrity and confidentiality of critical information, providing a robust defense against various attack vectors. The ongoing research and development in this field will continue to drive innovation and improve the effectiveness of hardware-based security solutions, ensuring that PUF-ECC integration remains at the forefront of secure technology advancements.

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